

Week 8

Three Pillars of Machine Learning

Linear Algebra

Probability

Optimization

Gradient Decent

Update Rule

Taylor Series

Newton's Method

Update Rule

Three Pillars of Machine Learning

Linear Algebra

Structure/ Relationship between data

Probability

Noise/ Uncertainty

Optimization

Data → Decisions

Gradient Decent

- converges to local minimum
- In gradient descent, we want to choose a direction d such that $f(x + \eta d) - f(x) < 0$

Update Rule

$$x_{t+1} = x_t - \eta f'(x_t)$$

Taylor Series

Using Taylor's series expansion we can obtain global information using local information of a function.

$$f(x + h) = f(x) + hf'(x) + \frac{h^2}{2}f''(x)$$

Newton's Method

Update Rule

$$x_{t+1} = x_t - \frac{f'(x_n)}{f''(x_n)}$$